

Software–Hardware Cooptimized High-Throughput CNN Acceleration Architecture for On-Board Remote Sensing Based on Distribution Difference-Aware Mixed-Precision Quantization

Haitao Chen , He Chen , Ning Zhang , Tong Wang , Shuo Ni, and Liang Chen 

I. INTRODUCTION

Abstract—Convolutional neural networks (CNNs) have demonstrated outstanding performance in various remote sensing intelligent tasks in recent years. However, in scenarios such as on-board emergency response, second-level on-board processing is required to achieve both low latency and high accuracy, while the limited resources and power constraints of on-board platforms make it difficult to process the large-scale parameters and intensive computation of CNNs efficiently on-chip. To address the aforementioned issues, this article proposes a real-time on-board CNN acceleration architecture based on distribution difference-aware mixed-precision quantization (DDAMQ), achieving efficient deployment through collaborative hardware and software optimization. At the algorithm level, low-bit quantization is employed to reduce computational and resource overhead. Furthermore, feature distribution difference modeling and outlier suppression are used to adaptively optimize the quantization bit width and pruning threshold, significantly mitigating the accuracy loss caused by low-bit quantization and improving model performance. At the hardware level, the computation and storage structure is optimized to improve resource utilization and reduce overall power consumption, making this architecture more suitable for resource-constrained real-time intelligent processing tasks. A resource consumption evaluation model is constructed to provide guidance for efficient deployment that meets the on-chip resource constraints of field-programmable gate arrays. Experiments based on four mainstream CNN models on five typical datasets are conducted and validated on an AMD-Xilinx VC709 development board. The results show that, at a similar compression ratio, the proposed DDAMQ strategy achieves superior classification performance; compared with recent state-of-the-art work, the designed CNN acceleration architecture improves throughput and energy efficiency by at least 2.17 and 2.13 times.

Index Terms—Bit width allocation, convolutional neural network (CNN), field-programmable gate array (FPGA)-based accelerator, low bit-width quantization, software–hardware cooptimization.

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Haitao Chen, He Chen, Tong Wang, Shuo Ni, and Liang Chen are with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 100081, China (e-mail: 3220240909@bit.edu.cn).

Ning Zhang is with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 100081, China, and also with the Department of Computing, HongKong Polytechnic University, Hong Kong (e-mail: nzhang.rs@bit.edu.cn).

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IN ON-BOARD remote sensing missions, satellites need to process massive amounts of data under limited downlink bandwidth and strict response time constraints, which leads to significant transmission delays. Convolutional neural networks (CNNs) are widely used in ground remote sensing image interpretation tasks such as image classification [1], object detection [2], and semantic segmentation [3] due to their powerful end-to-end feature representation capabilities. However, CNNs typically contain tens or even hundreds of millions of parameters and require a large amount of computation [4], making them difficult to deploy on resource-constrained remote sensing edge devices. Therefore, achieving efficient on-board CNN processing can significantly reduce data backhaul pressure, shorten processing links, and improve mission response speed, which is particularly important for applications such as disaster monitoring, emergency rescue, and rapid target identification.

To reduce the cost of deploying CNNs on on-board edge devices, existing studies have proposed numerous neural network model compression methods, including low-rank decomposition [5], pruning [6], knowledge distillation [7], and quantization [8]. Among these compression methods, quantization is an effective technique that represents weights and activations with low-bit precision, which offers several advantages for hardware deployment of network models. A key advantage is that it significantly reduces storage overhead and computational cost and hardware mapping area during inference on on-board edge devices, thereby lowering power consumption and processing latency [9]. However, low-bit representation often leads to severe performance degradation [10]. To address performance degradation, quantization-aware training (QAT) has become a key optimization strategy [11]. Based on quantization granularity, QAT is classified into hierarchical, group-level, and channel-level types. By integrating mixed-precision [12] and nonlinear quantization [13] at different granularities, QAT effectively mitigates accuracy loss. In neural networks, features in different layers often have different intraclass and interclass differences: the former reflects the difference in the average feature distribution of input samples of the same class, while the latter measures the difference in the average feature distribution of input samples between different classes. However, these methods neglect the